ENTERPRISE HEALTHCARE RECONCILIATION | DEPLOYED LIVE ON AWS EC2

Stitch Financial — Claims Reconciliation & Audit Hub

Author: Nakul (Forward Deployed Engineer Candidate) | **Reviewer:** Senior Engineering Review (ServiceNow) | **Status:** 38/38 Tests Passing

Live AWS Cloud Deployment: http://67.202.41.44:3000

Hosted on AWS EC2 (us-east-1) with Dockerized MySQL 8.0 & FastAPI • GitHub: https://github.com/nakul143naps/claim-reconciliation

1. Executive Problem & Technical Scope

In healthcare revenue cycle management, medical clinics submit bills (**Claims**) to insurance payers and receive back complex payment advices (**Remittances**). These remittances bundle cash transfers, contractual write-offs, copays, and clinical denial codes across multiple dates.

Why traditional tools fail: Standard billing software suffers from floating-point rounding drift (e.g. 0.1 + 0.2 = 0.30000000000000004), loses cash accountability by discarding orphan checks, and conflates operational denials with mathematical balances. This platform solves all three using pure integer cents arithmetic (BIGINT), strict foreign-key isolation via the **Dual-Key Pattern**, and full multi-page scale pagination.

2. Full-Stack Production Architecture

Layer	Technologies Used	Key Architectural Rationale
Frontend	React 18.3, TypeScript 5.6, Vite 6, Tailwind CSS	Single-page app with single-line filter bar, live search, Show: 10/20/50 selector, and audit drawers.
Backend API	Python 3.13, FastAPI, SQLAlchemy 2.0, Pydantic v2	Asynchronous REST API with eager relationship preloading (selectinload) eliminating N+1 queries.
Database	MySQL 8.0 (Containerized in Docker)	Non-negative CHECK constraints, composite timeline indexes, and ON DELETE SET NULL orphan safety.
Cloud (AWS)	EC2 (t3.micro), 20 GB gp3 EBS, 2 GB Swap	Expanded EBS online to 20 GB without downtime; allocated 2 GB swap to eliminate 1 GB RAM bottlenecks.

3. The Live Settlement Ledger: Plain-English Concept Guide

When you view the dashboard, the top container renders the **3-Way Live Settlement Ledger**. Every figure adheres to: Difference = Billed - Paid - Adjustments - Patient Responsibility.

Total Billed: $1,700.00 Total gross clinic charges across all 12 claims.	**Matched Cash: $990.00 (58.2%)** Cash received into the bank against valid claims.
Gross Uncollected: $710.00 Charges billed minus matched cash received ($1,700 - $990).	**Net Unsettled AR: $495.00** Actual collectible variance after $140 write-offs & $75 copays.

• **5 Balanced Claims ($0 balance)**: Fully settled (C1001, C1002, C1007, C1008, C1012).
• **6 Unbalanced Claims**: Variance remaining ($435.00 total) requiring billing follow-up.
• **1 Unremitted Claim (C1005)**: Zero remittances received ($60.00 billed). Difference is strictly null so accountants don't falsely classify it as debt.
• **1 Orphan Remittance (R1012)**: $50.00 check pointing to non-existent claim C9999. Quarantined in the unmatched queue with claim_id = NULL without losing cash.

4. Interactive Review Checklist (Please Try on Live App!)

• **Step 1: Open the Live URL** • Visit http://67.202.41.44:3000 in Chrome or Edge. Test the entries selector (Show: 10 / 20 / 50) and live filters.

• **Step 2: Inspect Claim C1007** • Click row C1007. Notice it evaluates as **Balanced** ($0 balance) while preserving the **Denial History** badge. The timeline shows it was originally denied under CO-197 ($0 paid), then settled under R1007 ($110 paid, $40

write-off).

• **Step 3: Review Denial Playbook** • In the detail modal, observe the clinical appeal playbook with root cause analysis and a copyable formal reconsideration letter.

• **Step 4: Scale Test via CSV Upload** • Click **'Import CSV'** in the table header. Upload data/claims_large.csv (50 claims) and data/remittances_large.csv. Watch the ledger recalculate live to **$10,085.00 Billed**, **$7,285.00 Paid**, and page through 50 claims!

• **Step 5: 1-Click Benchmark Reset** • Click **'Import CSV'** ➔ click **'Reset to Demo Benchmark'**. The database restores the official 12-claim benchmark in milliseconds.